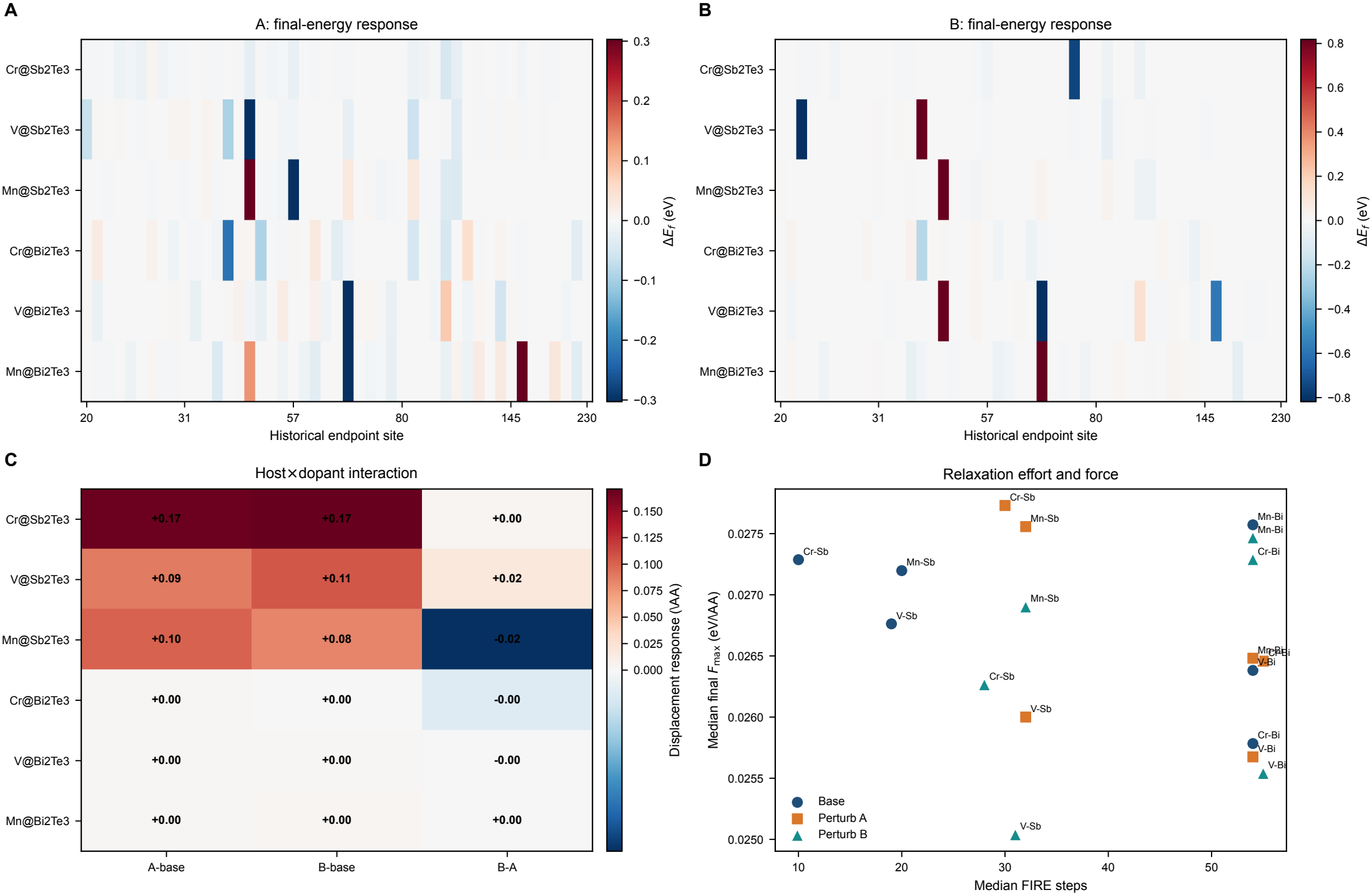

A
A: final-energy response
Cr@Sb2Te3
V@Sb2Te3
Mn@Sb2Te3
Cr@Bi2Te3
V@Bi2Te3
Mn@Bi2Te3
20
31
57
80
145
230
Historical endpoint site
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
ΔE_f (eV)
B
B: final-energy response
Cr@Sb2Te3
V@Sb2Te3
Mn@Sb2Te3
Cr@Bi2Te3
V@Bi2Te3
Mn@Bi2Te3
20
31
57
80
145
230
Historical endpoint site
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
ΔE_f (eV)
C
Host×dopant interaction
Cr@Sb2Te3
V@Sb2Te3
Mn@Sb2Te3
Cr@Bi2Te3
V@Bi2Te3
Mn@Bi2Te3
+0.17
+0.17
+0.00
+0.09
+0.11
+0.02
+0.10
+0.08
-0.02
+0.00
+0.00
-0.00
+0.00
+0.00
-0.00
+0.00
+0.00
+0.00
A-base
B-base
B-A
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Displacement response (ÅA)
D
Relaxation effort and force
Cr-Sb
Mn-Sb
Mn-Bi
Mn-Bi
Cr-Sb
Mn-Sb
Cr-Bi
Mn-Sb
V-Sb
Mn-Bi
Cr-Bi
V-Bi
Cr-Sb
V-Sb
Cr-Bi
V-Bi
V-Bi
V-Sb
Base
Perturb A
Perturb B
0.0275
0.0270
0.0265
0.0260
0.0255
0.0250
Median final F_max (eV/ÅA)
10
20
30
40
50
Median FIRE steps